

Total No. of Questions : 6]
P5038

SEAT No. :
[Total No. of Pages : 3

T.E./Insem. - 536
T.E. (Computer)
THEORY OF COMPUTATION
(2012 Pattern) (Semester - I)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates :

- 1) *Neat diagrams must be drawn wherever necessary.*
- 2) *Figures to the right side indicate full marks.*
- 3) *Assume Suitable data if necessary.*

Q1) a) Define Regular Expression. What are its limitations? List applications of RE. **[3]**

b) Define Pumping Lemma Using Pumping Lemma for regular sets prove the following :

$L = \{0^m 1^n 0^{m+n} \mid m \geq 1 \text{ and } n \geq 1\}$ is not regular **[4]**

c) Draw a FA for strings which do not contains 00 as substring over alphabets $\{0, 1\}$. **[3]**

OR

Q2) a) Show by principle of mathematical induction $1^2 + 2^2 + 3^2 + \dots + n^2 = n(n+1)(2n+1)/6$. **[4]**

b) Define Countable Set. Show that any subset of a countable set is countable. **[3]**

c) Define Deterministic and Non Deterministic Finite Automata in terms of 5-Tuple and discuss various applications of it. **[3]**

P.T.O.

- Q3)** a) Construct a DFA over the alphabets $\{0, 1\}$ for accepting the strings [4]
 i) Ending with 10
 ii) Ending with 11
 b) State and Explain Arden's theorem. [2]
 c) Convert the NFA shown in **Table 1** to its equivalent DFA. [4]

OR

- Q4)** a) List the limitations of finite automata. [2]
 b) Consider the Moore machine described by **Table 2**. Construct the corresponding Mealy Machine. [4]
 c) Design a NFA to recognize the strings wwyz, wxy, wyz, wxy over alphabets $\{w, x, y, z\}$ [4]

	0	1
$\rightarrow p$	$\{p, q\}$	$\{p\}$
q	$\{r\}$	$\{r\}$
r	$\{s\}$	$\{\emptyset\}$
s^*	$\{s\}$	$\{s\}$

Table 1

Present State	Next state		Output
	A=0	A=1	
Q1	Q1	Q2	0
Q2	Q1	Q3	0
Q3	Q1	Q3	1

Table 2

- Q5)** a) What is ambiguous grammar? Show that the grammar below is ambiguous, & find the equivalent unambiguous grammar. [4]
 $E \rightarrow E + E | E * E | (E) | I, I \rightarrow a | b$
 b) Define Context free Grammar & give its Applications. [2]
 c) Simplify the following grammar [4]
 $S \rightarrow 0A0 | 1B1 | BB, A \rightarrow C, B \rightarrow S | A, C \rightarrow S | \epsilon$

OR

- Q6)** a) Construct a DFA for the following left linear grammar [4]
 $S \rightarrow B1 \mid A0 \mid C0, B \rightarrow B1 \mid 1, A \rightarrow A1 \mid B1 \mid C0 \mid 0, C \rightarrow A0$
- b) Check whether the given grammar is in GNF. If not then find its equivalent GNF. [4]
 $S \rightarrow AA \mid a, A \rightarrow SS \mid b$
- c) Define Concurrent Grammar. [2]
